Title :Campus Rent A Full-Stack Web Application for item Renting in University Environment

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INTRODUCTION

- Global shift from ownership-based to access-based consumption.
- **Peer-to-peer rental platforms allow sharing of underutilized resources.**
- **University students often require items temporarily, such as:**

  Eg.:- Electronics, Lab Instruments, Sports Equipment, Furniture, e.t.c.
- **Purchasing items for short-term use:**

  Increases financial burden

  Leads to resource wastage
- **Campus Rent offers:**

  A secure, student-only rental marketplace

  Affordable and sustainable access to shared resources

  Verified and trusted campus-based transactions

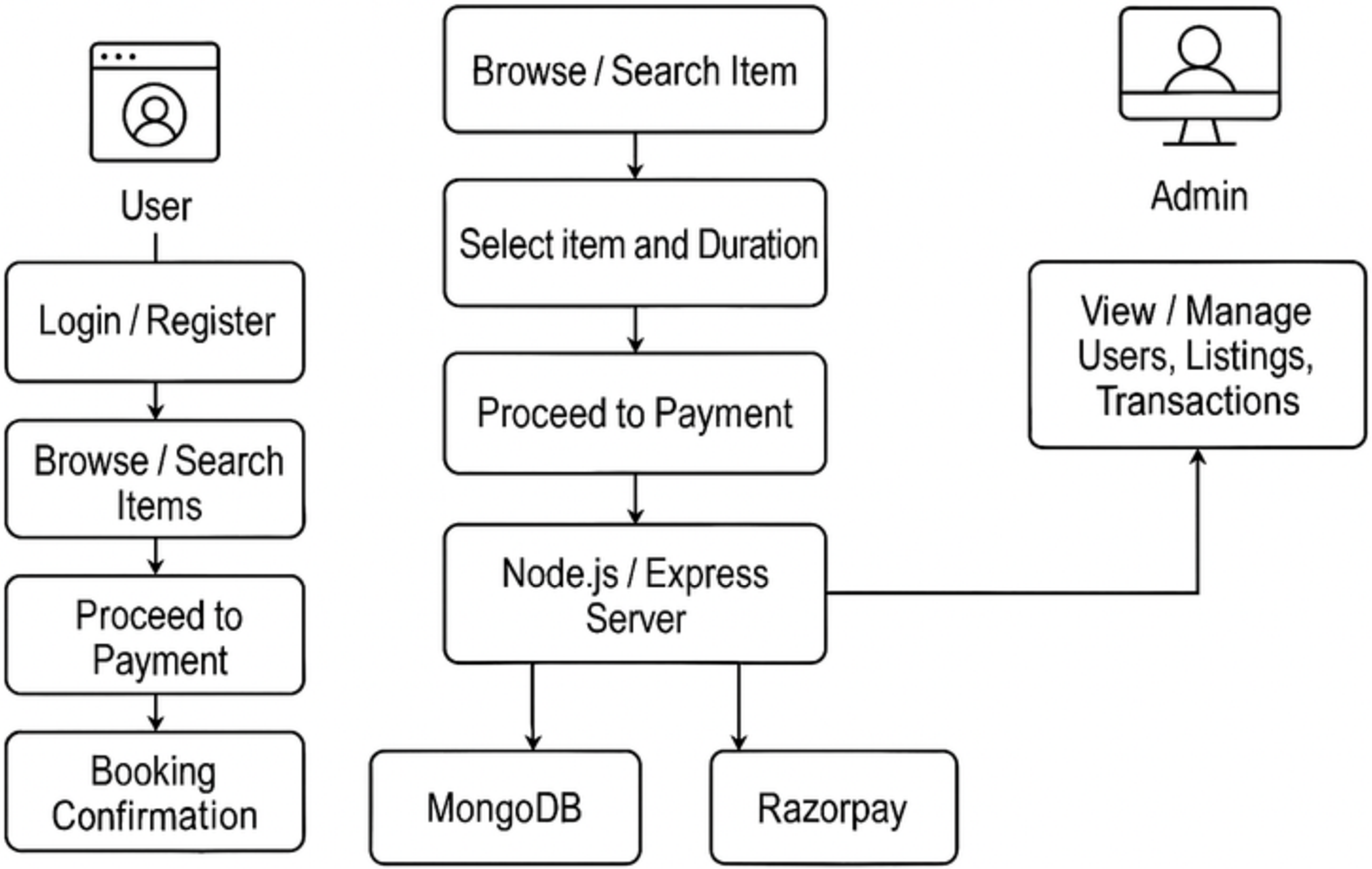
OBJECTIVES

- Develop a **centralized campus rental platform.**
- **Implement secure authentication**
  JWT-based login
  Encrypted password storage
- Enable **multi-category item listings.**
- Integrate **secure online payments** using Razorpay.
- Build a **scalable MERN full-stack system.**
- Promote sustainability through shared resource usage.
- Ensure **admin moderation,** transparency, and dispute handling.

PROPOSED SYSTEM MODEL

- **Client Layer (React.js Frontend):**
    - User login & registration
    - Browse rental categories
    - Item booking interface
    - Payment processing UI
- **Server Layer (Node.js + Express):**
    - API routing
    - Business logic handling
    - Booking workflow
    - Payment verification
- **Database Layer (MongoDB):**
    - Users collection
    - Items collection
    - Bookings collection
    - Transaction records
- **Payment Gateway (Razorpay):**
    - Simulate Order creation
    - Secure checkout
    - Payment signature verification

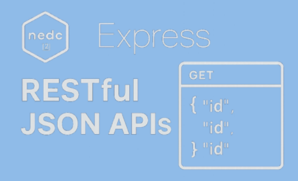
System Architecture Layers



IMPLEMENTATION

Frontend

- Developed using React.js
- Responsive design for:
  Mobile devices
  Laptops
- Category-wise item browsing
- Real-time booking interface



Backend

- Built with Node.js and Express
- RESTful JSON APIs
- Non-blocking and scalable architecture

IMPLEMENTATION

Security

- JWT-based authentication
- Bcrypt password hashing
- Razorpay payment signature verification

Database

- MongoDB with Mongoose ODM
- Flexible document-based schema
- Efficient data retrieval

RESULTS AND DISCUSSION

Functional Testing

All modules working successfully:

- Authentication
- Item posting
- Booking system
- Payment confirmation

Performance Analysis

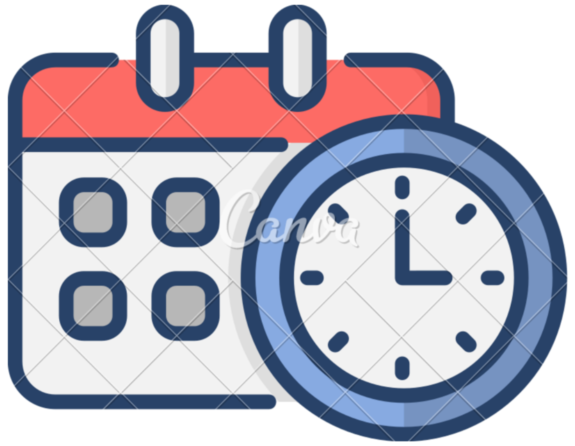
- Booking process completed in under 2 minutes
- System handled 200+ concurrent users
- Minor latency observed under peak load

User Feedback

- Pilot test conducted with 20 students
- Positive response received
- Increased trust and usability reported

Security Validation

- Encrypted password storage
- Secure Razorpay transaction processing



CONCLUSION AND FUTURE SCOPE

Conclusion

- Successfully developed a secure and scalable peer-to-peer rental platform
- Reduced unnecessary purchases among students
- Encouraged sustainability and resource sharing
- Improved trust through verified student-only access

Future Scope

- AI-based personalized recommendation system
- IoT-enabled smart lockers for item access
- Blockchain-based smart contracts
- Dedicated mobile application (Android/iOS)
- Advanced stress testing (50,000+ users)
- Enhanced usability and UX testing

Thank You...